

Jupsoft Link Management Platform (JLMP)

— PRD v1.0

Sep 28, 2026 · @Sachin Sharma

Executive summary

JLMP is a branded short-link, QR code and click-attribution layer for Jupsoft products, launched internally first and opened to paying customers only after it proves value. Its edge is not shortening links; it is connecting a click to a business result inside the Jupsoft ecosystem, such as a fee paid, an enquiry raised or an admission confirmed.

Item	Detail
Product	Jupsoft Link Management Platform (JLMP)
Document version	1.0 (draft for review)
Owner	Sachin, Director, Jupsoft Systems Pvt. Ltd.
Audience	Product, engineering, QA, DevOps, sales leadership
Stack	Node.js (TypeScript), PostgreSQL, Redis, Cloudflare
Release strategy	Internal MVP → internal rollout → commercial launch

The MVP ships multi-tenant from day one, so the commercial launch needs no redesign. It deliberately defers billing, white labelling and heavy analytics infrastructure until usage justifies them.

Problem statement

Jupsoft products send thousands of links to parents, students and prospects, but cannot show whether those links led to action. Today these links are long, unbranded or routed through third-party shorteners.

- **Cost and length:** long URLs push SMS past the 160-character segment limit, raising message cost.
- **Trust:** parents increasingly ignore generic short links (bit.ly, tinyurl) because they resemble phishing.

- **Compliance:** TRAI requires URLs in commercial SMS to be whitelisted on the DLT platform; third-party shortener domains complicate this.
- **No attribution:** schools cannot see whether a fee reminder led to a payment, or an admission campaign to an enquiry.
- **Dependency:** third-party services can change pricing, rate-limit, expire links or hold click data outside Jupsoft's control.

The opportunity is to turn every link Jupsoft products send into a measurable, trusted touchpoint, and to offer that capability to customers as part of the Jupsoft ecosystem.

Vision, positioning and objectives

JLMP's vision is to make every link an organization sends trusted, trackable and tied to an outcome. It is positioned as the communication and attribution layer of the Jupsoft ecosystem, not a standalone Bitly competitor.

Positioning statement: JLMP helps organizations send branded, secure links and QR codes, and see which messages actually drive payments, enquiries and admissions, by connecting link activity to the Jupsoft systems they already use.

Why this wins where generic shorteners don't: Bitly, Rebrandly, Short.io and open-source tools compete on price and features. JLMP competes on attribution inside Jupsoft ERP, Admission CRM, CRM, LMS and DigifyNext campaigns, which generic tools cannot see.

Objective type	Objective	Measured by
Business	Replace third-party shorteners across all Jupsoft products	Share of product-sent links on JLMP
Business	Add measurable value to existing ERP and CRM contracts	Customer usage of link reports
Business	Create a future SaaS revenue line	Paid tenants after commercial launch
Customer	Show institutions which communication drives action	Click-to-outcome reports in use
Technical	Multi-tenant, API-first from the MVP	No redesign needed for commercial launch
Technical	Fast, reliable redirects	Latency and uptime targets (see Non-functional requirements)

Target users and roles

The MVP serves Jupsoft's own products and their institutional customers; external businesses and agencies come with the commercial launch.

Segment	Examples	Phase	Primary need
Internal products	eConnect ERP, Admission CRM, CRM, LMS	MVP	Short links via API inside existing notifications
Education customers	Schools, colleges, universities, coaching institutes	MVP (via ERP)	Trusted fee, admission and circular links; outcome reports
Jupsoft marketing	DigifyNext campaigns	MVP	Campaign links, UTM tracking, QR codes
Businesses	SMEs, healthcare, manufacturing, NGOs, retail	Commercial	Branded links and campaign analytics
Agencies	Digital marketing and SEO agencies	Commercial	Multi-client link management, white label

Role	Can do
Platform Super Admin (Jupsoft)	Manage tenants, plans, domains, abuse reports, global settings
Tenant Admin	Manage users, links, campaigns, domains and API keys for one tenant
Manager	Create links and campaigns; view all tenant analytics
User	Create and manage own links; view own analytics
Read Only	View reports only

Scope by phase

The MVP is limited to what Jupsoft's own products need to replace third-party shorteners and show click-to-outcome data.

Capability	MVP	Phase 2	Phase 3
Tenant and user management with roles	Yes		
Short links (random code + tenant-scoped alias)	Yes		
Redirect service (302/307) with expiry and disable	Yes		
Malicious-URL screening and abuse reporting	Yes		
Static QR codes (PNG, SVG)	Yes		
Dynamic QR codes (editable destination)	Yes		
Basic analytics: clicks, unique clicks, device, referrer, date	Yes		
REST API with API keys	Yes		
Bulk link creation (API and CSV)	Yes		
Outcome attribution hooks (fee paid, enquiry created)	Yes		
Campaigns and UTM builder		Yes	
Tenant subdomains and customer custom domains		Yes	
Billing, plans and usage metering		Yes	

Capability	MVP	Phase 2	Phase 3
White label (logo, colours, branded reports)		Yes	
Report exports (CSV, Excel, PDF)		Yes	
SSO for enterprise tenants			Yes
Funnel reports (admission, fee collection)			Yes
AI campaign insights and send-time recommendations			Yes

Out of scope for all phases in this PRD: link-in-bio pages, digital business cards, and a public self-service free tier. Each can be proposed separately once the core platform is proven.

Functional requirements

The MVP has 16 functional requirements; each must pass its acceptance criteria before internal rollout.

ID	Requirement	Acceptance criteria	Phase
FR-01	Create tenant	Super Admin creates, suspends and reactivates a tenant; a suspended tenant's links show a neutral "link unavailable" page	MVP
FR-02	Manage users and roles	Tenant Admin invites users and assigns one of five roles; every action checks role and tenant	MVP
FR-03	Create short link	Input: destination URL, optional alias, expiry, max clicks, tag. Output: short URL and QR code in under 500 ms	MVP
FR-04	Short code generation	Random base62 code, 6–7 characters, excluding look-alikes (0/O, 1/I/l); unique constraint in database	MVP
FR-05	Tenant-scoped alias	Readable aliases are unique within a tenant only, served under a tenant prefix (e.g. /hw/admission-2027); reserved words blocked	MVP
FR-06	Redirect	Active link redirects with HTTP 302 by default (307 optional); 301 is not offered	MVP

ID	Requirement	Acceptance criteria	Phase
FR-07	Link lifecycle	Links can be edited, disabled, archived, cloned; expired or max-click links show a tenant-branded expiry page	MVP
FR-08	Destination screening	Every destination is checked against a malicious-URL service at creation and re-checked periodically; flagged links are blocked	MVP
FR-09	Abuse reporting	Public "report this link" page; reports queue for Super Admin review	MVP
FR-10	QR codes	Every link gets a QR code downloadable as PNG and SVG; dynamic QR codes keep the same code when the destination changes	MVP
FR-11	Click logging	Each click records timestamp, link, tenant, device type, browser, OS, referrer, country (from IP) and a hashed visitor key; raw IP is not stored	MVP
FR-12	Analytics dashboard	Per-link and per-tenant totals, unique clicks, device, referrer and daily trend over a chosen date range	MVP
FR-13	REST API	Every UI action is available via versioned API authenticated with tenant API keys	MVP
FR-14	API keys	Create, name, scope, rotate and revoke keys; last-used time shown	MVP
FR-15	Bulk creation	Up to 10,000 links per API batch or CSV upload, processed asynchronously with a status endpoint	MVP
FR-16	Outcome attribution	Products can attach an external reference (e.g. fee invoice ID) to a link and post an outcome event back; reports show clicks versus outcomes	MVP
FR-17	Campaigns and UTM	Group links into campaigns; UTM builder appends parameters; campaign-level reports	Phase 2
FR-18	Custom domains	Tenant adds a domain, verifies via DNS record, SSL issued automatically, health monitored	Phase 2
FR-19	Billing and metering	Plans, subscriptions, invoices; metering of links, clicks, API calls and users	Phase 2

ID	Requirement	Acceptance criteria	Phase
FR-20	White label	Tenant logo, colours and domain on dashboard, expiry pages and reports	Phase 2
FR-21	Exports	Reports exported as CSV, Excel and PDF	Phase 2

FR-16 is the requirement that differentiates JLMP; it should be designed with the ERP and Admission CRM teams before development starts.

Domain strategy and SMS compliance

JLMP uses two dedicated short domains, separate from jupsoft.com, so an abuse flag on public traffic can never break school fee links or company email.

Domain	Used for	Phase
Internal short domain	Links sent by Jupsoft products (ERP, Admission CRM, CRM, LMS) and DigifyNext	MVP
Public short domain	Links created by external commercial tenants	Commercial launch
Tenant subdomains	Readable branded links, e.g. hillwoods.<domain>/admission-2027	Phase 2
Customer custom domains	Premium tenants' own domains, e.g. go.school.edu.in	Phase 2

Domain selection rules

- 5–7 characters before the TLD, easy to read aloud and type from a printed circular.
- Recognizably connected to Jupsoft or its purpose, so recipients trust it.
- Prefer `.in` : short, stable under NIXI, familiar to Indian users. Avoid spam-associated TLDs (.xyz, .top, .tk) and foreign ccTLDs with unpredictable registry rules.
- Registrar lock, auto-renewal, DNSSEC and Cloudflare in front, from day one.

SMS DLT compliance

- Each short domain must be whitelisted on the DLT platform before use in SMS, for Jupsoft and for each sender entity (school) where required.
- SMS templates must reference the whitelisted domain; the SMS vendor must confirm the exact whitelisting process before the MVP pilot.

- The admin console will show each tenant's DLT whitelisting status for its domains.

Alias collisions are avoided by scoping readable aliases to a tenant prefix in the MVP and to tenant subdomains in Phase 2. Random codes remain the default.

Non-functional requirements

Targets are set for realistic MVP volumes and reviewed once internal usage data exists.

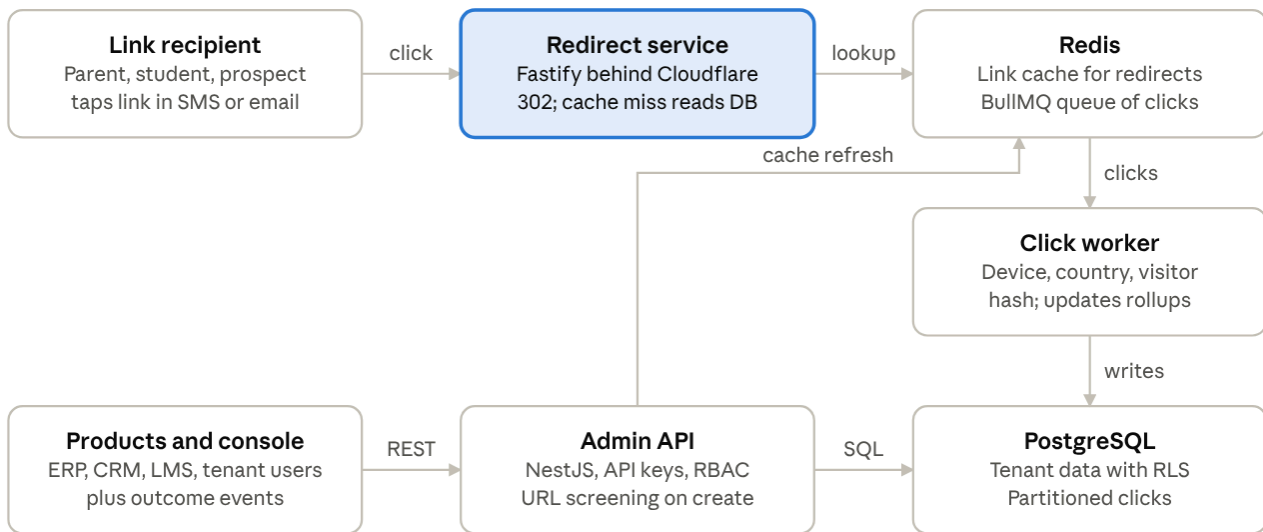
Area	MVP target	Later target
Redirect latency (server, p95)	Under 50 ms on cache hit, under 150 ms on miss	Under 30 ms at edge
API latency (p95)	Under 500 ms for single operations	Same
Availability, redirect service	99.9% monthly	99.95%
Availability, admin and API	99.5% monthly	99.9%
Capacity	1 million active links; 5 million clicks per month	Re-planned from real growth
Click logging	Never delays a redirect; asynchronous, at-least-once	Same
Data retention	Raw clicks 13 months; daily aggregates kept for tenant lifetime	Configurable per plan
Recovery	Daily backups, point-in-time recovery; RPO 1 hour, RTO 4 hours	RPO 15 minutes
Browser support	Last two versions of Chrome, Edge, Safari, Firefox; mobile responsive	Same
Accessibility	Admin console meets WCAG 2.1 AA for core flows	Same

If the redirect service fails, links already cached at the edge should keep resolving; this is a design goal for Phase 2.

Architecture and technology stack

JLMP runs as two Node.js services on a shared PostgreSQL and Redis base: a lean redirect service on the hot path and a full admin API for everything else.

Redirects never wait on the database write



JLMP architecture · redirect path and admin path

A click is answered from the Redis cache and logged through a queue, so database load never slows a redirect. The admin API refreshes the cache whenever a link changes.

Layer	Choice	Reason
Language	TypeScript only (Node.js LTS)	Type safety across both services; no plain JavaScript
Redirect service	Fastify	Minimal overhead on the latency-critical path
Admin API	NestJS	Modules, guards and dependency injection suit multi-tenant RBAC
Web console	React-based SPA (framework to confirm)	Tenant dashboard, analytics, link management
Database	PostgreSQL with Row-Level Security	Relational core; RLS as a second line of tenant isolation
Data access	Drizzle or Prisma (team to choose)	Typed queries and migrations
Cache and queue	Redis with BullMQ	One system for link cache and click queue; replaces RabbitMQ
Edge	Cloudflare (DNS, TLS, WAF, rate limiting)	DDoS protection and bot filtering in front of redirects

Layer	Choice	Reason
Hosting	AWS containers (ECS)	Consistent with existing AWS usage; Kubernetes not needed at MVP
Observability	Structured logs, metrics and alerts on redirect latency and queue depth	Early warning before users notice

Deliberately deferred: ClickHouse, RabbitMQ and Kubernetes. They are reconsidered only if monthly clicks outgrow partitioned PostgreSQL.

Data model

Every tenant-owned table carries `tenant_id`, and PostgreSQL Row-Level Security enforces it even if application code misses a filter.

Table	Key fields	Notes
tenants	id, code, name, status, plan_id, created_at	Code doubles as MVP alias prefix (e.g. hw)
users	id, tenant_id, name, email, role, status, last_login_at	Five roles; Super Admins have no tenant_id
domains	id, tenant_id, hostname, type, verification_status, dlt_status	Internal, public, subdomain or custom
links	id, tenant_id, domain_id, short_code, alias, destination_url, redirect_type, status, expires_at, max_clicks, tag, external_ref, created_by, created_at	Unique (domain_id, short_code); unique (tenant_id, alias)
qr_codes	id, link_id, style, is_dynamic, created_at	Files generated on demand, cached in object storage
clicks	id, link_id, tenant_id, clicked_at, visitor_hash, device, browser, os, country, referrer, is_bot	Partitioned by month; raw IP never stored
click_daily	tenant_id, link_id, date, clicks, unique_clicks, by_device, by_country, by_referrer	Rollups used by all dashboards

Table	Key fields	Notes
outcomes	id, tenant_id, link_id, external_ref, outcome_type, value, occurred_at	Posted by Jupsoft products (FR-16)
api_keys	id, tenant_id, name, key_hash, scopes, last_used_at, revoked_at	Only the hash is stored
screening_results	id, link_id, provider, verdict, checked_at	Supports blocking and audit
abuse_reports	id, link_id, reporter_contact, reason, status, created_at	Reviewed by Super Admin
audit_log	id, tenant_id, actor_id, action, entity, before, after, created_at	All admin and API changes

Phase 2 adds plans, subscriptions, invoices, usage_counters, campaigns and tenant_branding without changing the tables above.

API overview

The REST API is versioned under `/api/v1`, authenticated with an `X-API-Key` header, and documented as an OpenAPI specification generated from code.

Method	Endpoint	Purpose
POST	<code>/api/v1/links</code>	Create a short link (returns short URL and QR URL)
POST	<code>/api/v1/links/bulk</code>	Create up to 10,000 links asynchronously
GET	<code>/api/v1/links/bulk/{jobId}</code>	Check bulk job status and download results
GET	<code>/api/v1/links</code>	List and search links
GET	<code>/api/v1/links/{id}</code>	Get one link
PATCH	<code>/api/v1/links/{id}</code>	Update destination, expiry, status or tag
DELETE	<code>/api/v1/links/{id}</code>	Archive a link
GET	<code>/api/v1/links/{id}/qr</code>	Download QR as PNG or SVG
GET	<code>/api/v1/links/{id}/analytics</code>	Clicks and breakdowns for a date range
GET	<code>/api/v1/analytics</code>	Tenant-level analytics

Method	Endpoint	Purpose
POST	/api/v1/outcomes	Record an outcome against a link or external reference
GET/POST/DELETE	/api/v1/api-keys	Manage API keys (Tenant Admin only)

Conventions: idempotency keys on create endpoints, cursor pagination, per-key rate limits (starting at 60 requests per minute, raised for internal products), and consistent error codes. Signed webhooks for link events are a Phase 2 item.

Security, abuse prevention and privacy

The biggest operational risk is a short domain being blacklisted for phishing, so abuse controls are MVP requirements, not later additions.

Security

- HTTPS only with HSTS; TLS terminated at Cloudflare and re-encrypted to origin.
- Tenant isolation in application code plus PostgreSQL Row-Level Security.
- API keys stored as hashes, scoped, rotatable; admin console with strong passwords and optional two-factor authentication.
- Rate limiting at the edge and per API key; WAF rules on the admin and API hosts.
- Audit log for every admin and API change; dependency scanning in CI; annual third-party penetration test before commercial launch.

Abuse prevention

- Destination URLs screened against a malicious-URL service on creation and re-checked on a schedule.
- Block redirects to other shorteners (no chaining) and to known phishing patterns.
- Public "report this link" page; Super Admin can disable a link or suspend a tenant in one action.
- Internal and public traffic on separate domains so abuse by external tenants cannot affect school links.
- New commercial tenants start with low limits until verified.

Privacy (Digital Personal Data Protection Act, 2023)

- Default analytics are aggregate: counts by device, country, referrer and date. Raw IP addresses are not stored.

- Per-recipient tracking (for example, which parent opened a fee link) is only available when the tenant sends a recipient-specific link from a Jupsoft product and has given notice to recipients. This must be reviewed by legal counsel before release, including rules for children's data.
- Raw click data deleted after 13 months; tenants can request deletion of their data.
- Data hosted in India (AWS Mumbai region).

Success metrics

The commercial launch is gated on internal adoption and a measurable outcome improvement, not on link volume alone. Baselines marked "to measure" must be captured before the MVP pilot starts.

Metric	Baseline	Target	By
Share of links sent by Jupsoft products that use JLMP	0%	100%	End of internal rollout
Pilot schools using link reports monthly	0	At least 5	End of pilot
SMS segments per fee reminder	To measure	Reduced to 1 where message allows	End of pilot
Fee-reminder click-through rate	To measure	Measurable uplift versus baseline	End of pilot
Fee payments attributed to reminder links	Not possible today	Reported for every pilot school	End of pilot
Redirect latency (p95, server)	n/a	Under 50 ms cache hit	Continuous
Redirect uptime	n/a	99.9% monthly	Continuous
Domain blacklisting incidents	n/a	Zero	Continuous
Paid external tenants	0	Set at commercial launch decision	Commercial phase

Risks, assumptions and open questions

The two risks most likely to derail JLMP are DLT whitelisting delays and team capability on the new Node.js stack; both must be resolved before development starts.

Risk	Impact	Likelihood	Mitigation
DLT whitelisting slow or rejected for the short domain	SMS links blocked	Medium	Start whitelisting with the SMS vendor during domain purchase
Short domain blacklisted for abuse	All links on that domain fail	Medium	Separate internal and public domains; screening; fast takedown
Limited Node.js/TypeScript experience in a .NET-heavy team	Slower delivery, weaker code	Medium	Confirm team skills; appoint a Node lead; code review standards
Privacy non-compliance on per-recipient tracking	Legal and reputational exposure	Medium	Aggregate by default; legal review before release
Crowded shortener market limits commercial uptake	Revenue below plan	High	Compete on ecosystem attribution, not standalone features
Scope creep toward a full marketing suite	MVP delay	Medium	Hold to the phase table; new ideas go to later phases

Assumptions

- Jupsoft products can call the JLMP API from their notification flows.
- The ERP can post fee-payment events with a reference that matches the link.
- Pilot schools agree to share baseline communication data.

Open questions

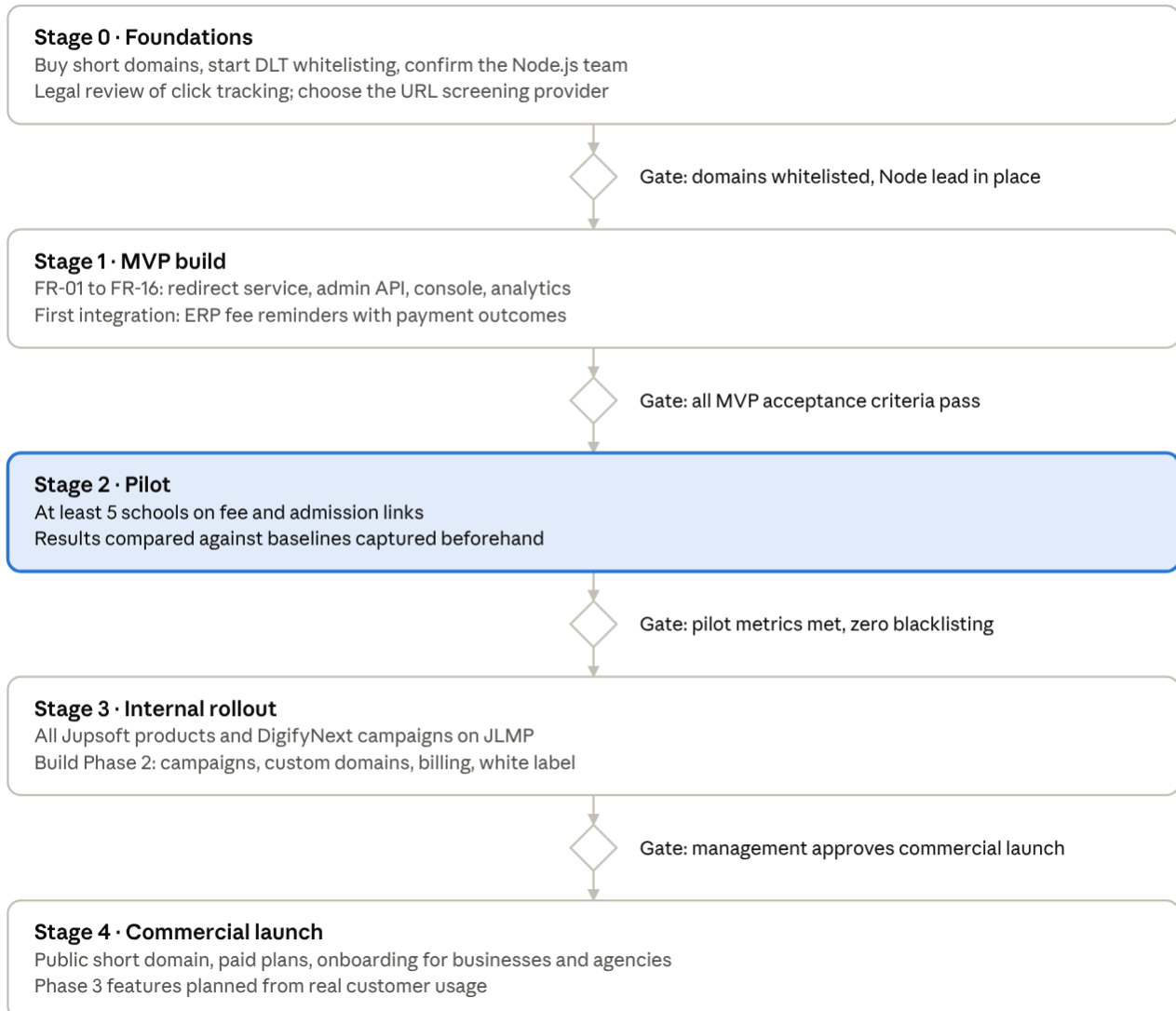
- ☐ Which short domain names are available, and which are chosen for internal and public use?
- ☐ Which Jupsoft products and product names are integrated first (confirm current names, e.g. Admission CRM, LMS)?
- ☐ Does the team have Node.js/TypeScript developers, and who leads?
- ☐ Web console framework: confirm React-based choice.
- ☐ Which malicious-URL screening provider and at what cost?
- ☐ What is the SMS vendor's exact DLT process for new domains?
- ☐ Which schools join the pilot?

- ☐ What pricing model will be tested at commercial launch?

Roadmap and milestones

JLMP is delivered in five stages, and no stage starts until the gate before it is passed. Durations and dates will be set after the team is confirmed and the effort estimate is done.

Commercial launch waits for proof from the pilot



JLMP delivery roadmap • 5 stages, 4 gates

The pilot is the decisive stage: it is where JLMP must show that links lead to payments and enquiries, which is the case for everything after it.